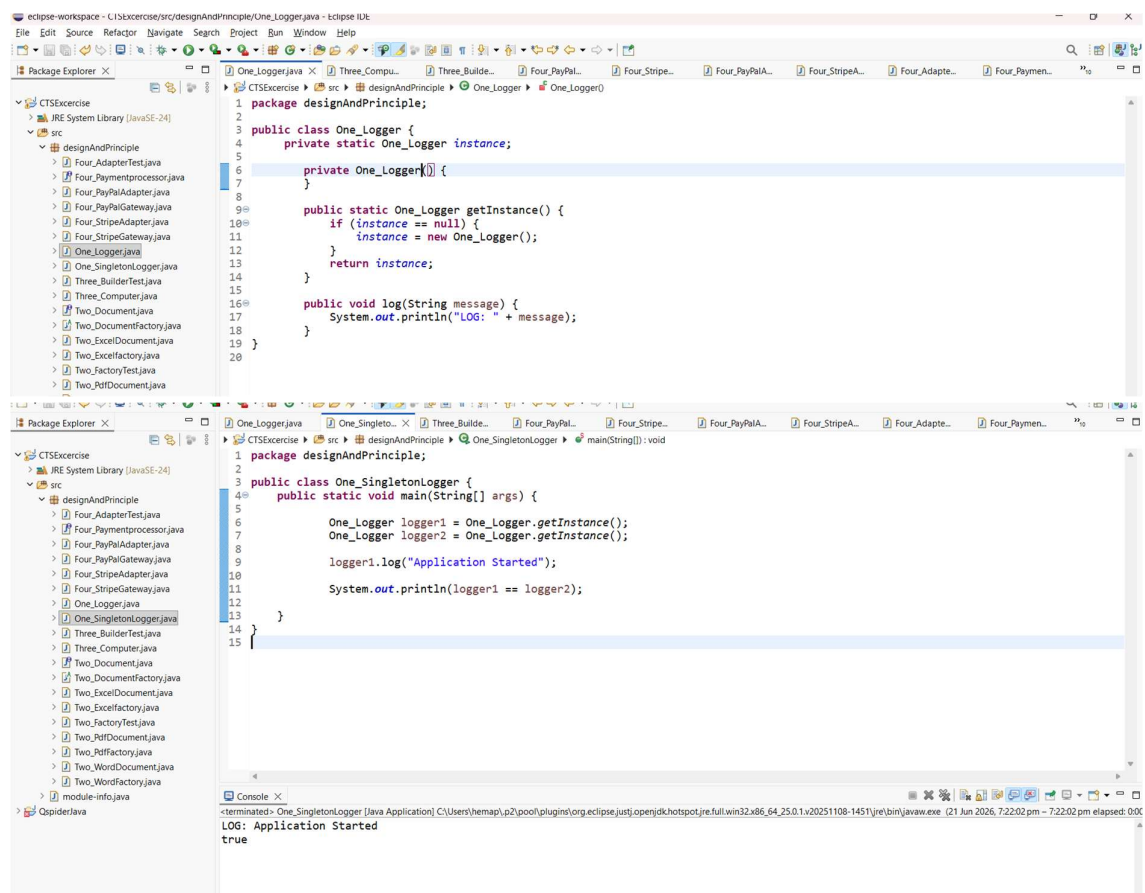


Exercise 1: Singleton Pattern

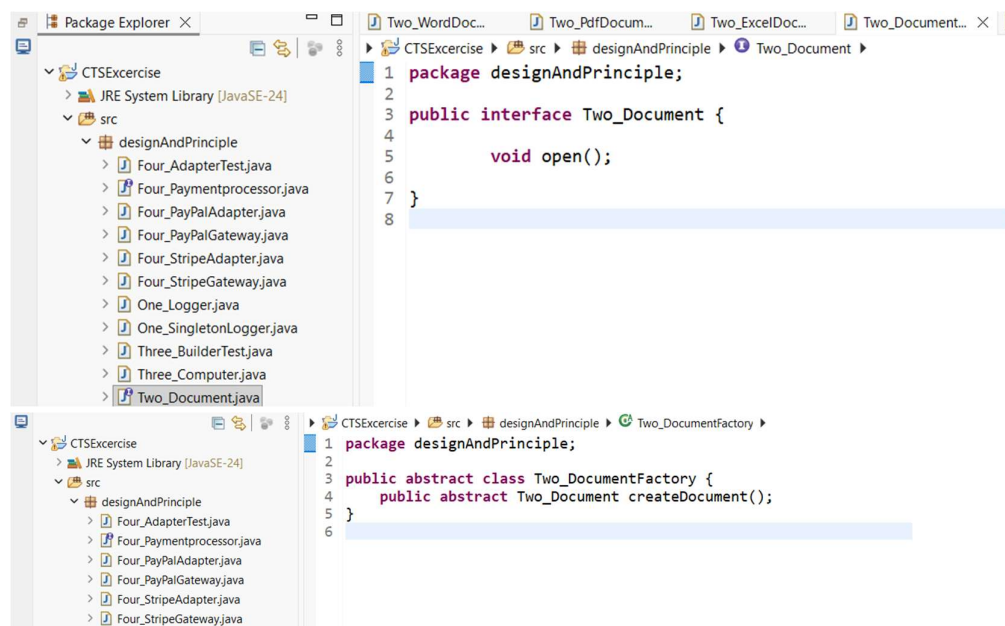


```
1 package designAndPrinciple;
2
3 public class One_Logger {
4     private static One_Logger instance;
5
6     private One_Logger() {}
7
8
9     public static One_Logger getInstance() {
10         if (instance == null) {
11             instance = new One_Logger();
12         }
13         return instance;
14     }
15
16     public void log(String message) {
17         System.out.println("LOG: " + message);
18     }
19 }
20
```

```
1 package designAndPrinciple;
2
3 public class One_SingletonLogger {
4     public static void main(String[] args) {
5
6         One_Logger logger1 = One_Logger.getInstance();
7         One_Logger logger2 = One_Logger.getInstance();
8
9         logger1.log("Application Started");
10
11         System.out.println(logger1 == logger2);
12     }
13 }
14
15
```

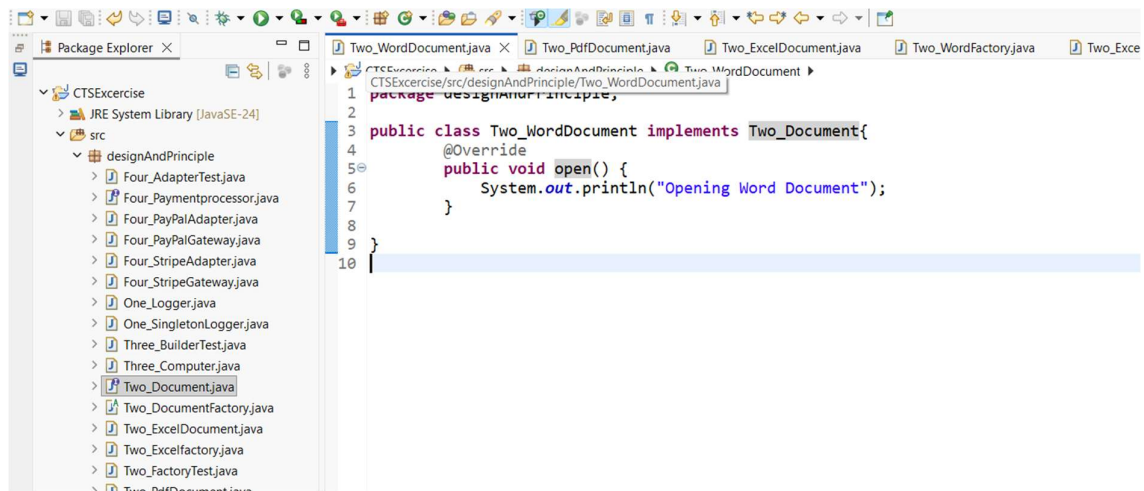
```
<terminated>: One_SingletonLogger [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk hotspot\jre.full.win32.x86_64_25.0.1.v20251108-1451\jre\bin\javaw.exe (21 Jun 2026, 7:22:02 pm - 7:22:02 pm elapsed: 0:00)
LOG: Application Started
true
```

Exercise 2: Factory Method Pattern

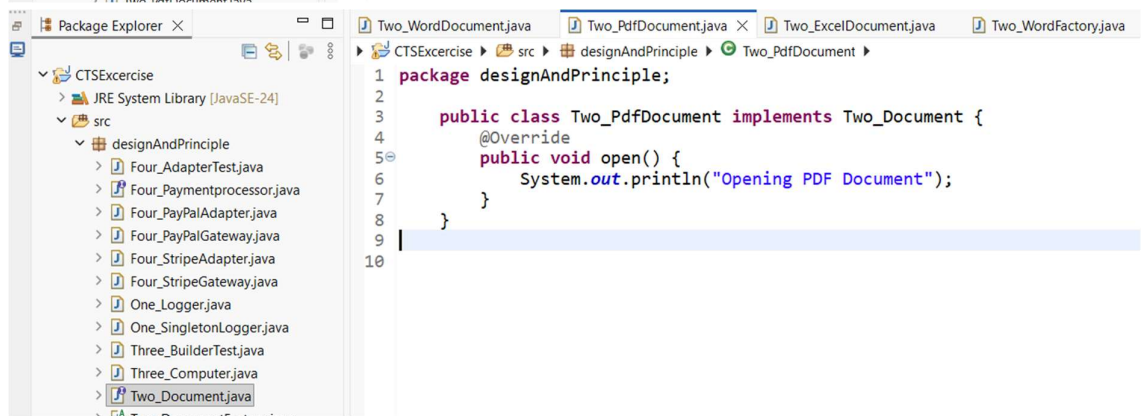


```
1 package designAndPrinciple;
2
3 public interface Two_Document {
4
5     void open();
6
7 }
8
```

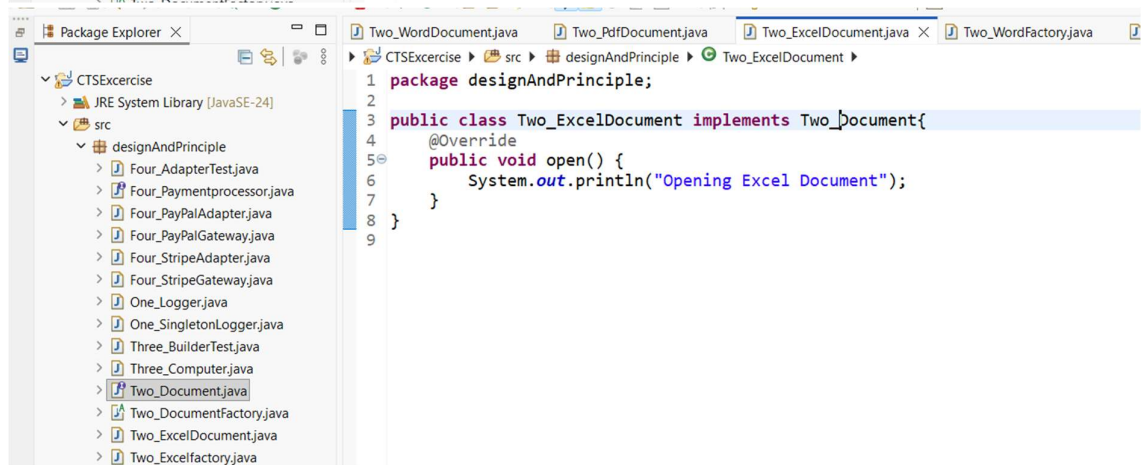
```
1 package designAndPrinciple;
2
3 public abstract class Two_DocumentFactory {
4     public abstract Two_Document createDocument();
5 }
6
```



```
1 package designAndPrinciple;
2
3 public class Two_WordDocument implements Two_Document{
4     @Override
5     public void open() {
6         System.out.println("Opening Word Document");
7     }
8 }
9
10
```



```
1 package designAndPrinciple;
2
3 public class Two_PdfDocument implements Two_Document {
4     @Override
5     public void open() {
6         System.out.println("Opening PDF Document");
7     }
8 }
9
10
```



```
1 package designAndPrinciple;
2
3 public class Two_ExcelDocument implements Two_Document{
4     @Override
5     public void open() {
6         System.out.println("Opening Excel Document");
7     }
8 }
9
```

